

Photovoltaic Inverter Modbus Serial Communications Protocol Specification

Revision History

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V2.00	31 Aug 2012	1. Added hardware definition. 2. Added Modbus configuration. 3. Added calibration formula. 4. Added power management description 5. Added error code table. 6. Added 3 phase monitoring parameters. 7. Added SunSpec Alliance Specification - Common Elements v1.4. 8. Added SunSpec Alliance Specification - Inverter Models v1.1. 9. Modified Discrete Inputs. 10. Modified the protocol to Word file.	Eric Lee
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V2.10	12 Mar 2013	1. Define the priority of error code 2. Added 0x086D,0x086E (Function control Flags) 3. Added 0x089F (R leakage gain)	Eric Lee
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Contents

1.	Introduction.....	- 5 -
2.	Related documents.....	- 5 -
3.	Hardware	- 6 -
3.1.	Cables	- 6 -
3.2.	Pin Lay-out	- 6 -
4.	Modbus Configuration.....	- 7 -
4.1.	Supported Transmission Medium	- 7 -
4.2.	Supported Modes of Transmission	- 7 -
4.3.	Communications Parameters	- 7 -
4.4.	Exception Responses	- 7 -
4.5.	CRC Generation	- 8 -
4.6.	Addressing Rules.....	- 8 -
4.7.	RTU Framing.....	- 8 -
5.	Supported Modbus Functions	- 10 -
5.1.	Function 1 (01h): Read Coils.....	- 10 -
5.2.	Function 2 (02h): Read Discrete Inputs	- 10 -
5.3.	Function 3 (03h): Read Holding Registers	- 11 -
5.4.	Function 4 (04h): Read Input Registers	- 11 -
5.5.	Function 5 (05h): Write Single Coil	- 12 -
5.6.	Function 6 (06h): Write Single Register	- 12 -
5.7.	Function 16 (10h): Write Multiple Registers	- 13 -
6.	CyberPower Systems Modbus Register Map	- 14 -
6.1.	CyberPower Systems Modbus Data Formats	- 14 -
6.2.	Input Registers	- 16 -
6.2.1.	Monitoring Parameters (Single Phase)	- 16 -
6.2.2.	Monitoring Parameters (Three Phase).....	- 19 -
6.2.3.	kWh & Runtime Logs (Single phase).....	- 22 -
6.2.4.	kWh & Runtime Logs (Three phase).....	- 22 -
6.2.5.	Event Log (Single phase, Refer to Appendix C for more detail).....	- 24 -
6.2.6.	Event Log (Three phase, Refer to Appendix C for more detail)	- 24 -
	Appendix A: Data Structure	- 26 -
	Appendix B: Grid codes and Languages Table	- 28 -
1.	Countries Table (Single phase).....	- 28 -
2.	Countries Table (Three phase)	- 28 -
3.	Languages Table (Single phase).....	- 29 -
4.	Languages Table (Three phase)	- 30 -
	Appendix C: Error Code	- 31 -
1.	Error Code Format for Single Phase	- 31 -
2.	Error Code Priority for Single Phase	- 31 -
3.	"E" Error Code Table for Single Phase	- 31 -
4.	"W" Error Code Table for Single Phase	- 32 -
5.	"F" Error Code Table for Single Phase	- 32 -
6.	Error Code Table for Three Phase	- 35 -

1. Introduction

This document explains the implementation of the Modbus communications protocol specific to CyberPower Systems photovoltaic inverters. The Modbus protocol is an industry standard that allows a master station to interrogate devices for its publically available data and set supported control and configuration parameters.

2. Related documents

<http://www.modbus.org/> Modicon Modbus Protocol Reference Guide
<http://www.modbus.org/> Modbus over Serial Line Specification and Implementation Guide, V1.2
<http://www.modbus.org/> Modbus Application Protocol Specification, V1.1b
<http://www.sunspec.org/> SunSpec Alliance Specifications

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3. Hardware

3.1. Cables

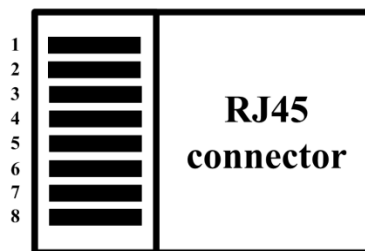
For the cabling CAT-5, ISDN or CPS Data Cable can be used. The maximum length of the cable is 1000m, but it is recommended not to exceed 500m.

3.2. Pin Lay-out

An RJ45 connector (8-pole telephone jack) is used to connect the various devices to the bus. The connector has the following pinout:

Single phase

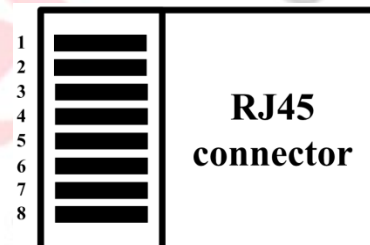
- 1 = - supply / ground
- 2 = + supply (12V)
- 3 = B
- 4 = A
- 5 = A
- 6 = B
- 7 = + supply (12V)
- 8 = - supply / ground



- 12V power supply is also available, which makes it possible to supply power to interfaces and/or displays.
- 4-pole cable can be used if the 12V power supply is not used.

Three phase

- 1 = N/A
- 2 = N/A
- 3 = B
- 4 = A
- 5 = N/A
- 6 = N/A
- 7 = N/A
- 8 = N/A



- Three phase doesn't supply 12V power to other interfaces and/or displays.

4. Modbus Configuration

4.1. Supported Transmission Medium

Products supporting Modbus communicate primarily via the RS-485 (TIA/EIA-485-A) communications standard. The RS-485 medium allows multiple devices to communicate on a single medium using a master/slave approach for arbitrating the bus. All CyberPower Systems products are 1/8 unit load (UL) devices. Depending on the mix of other devices from other vendors, biasing or polarization of devices on the network, this will allow for as few as 32 devices or as many as 255 devices on a single bus.

4.2. Supported Modes of Transmission

The Modbus protocol supports two serial transmission modes, ASCII and RTU (Remote Terminal Unit). CyberPower Systems only supports the RTU mode of operation.

4.3. Communications Parameters

Serial communications default to 8 data bits, no parity, and 1 stop bit. A default baud rate of 9600 BPS is used. Parameters are reconfigurable.

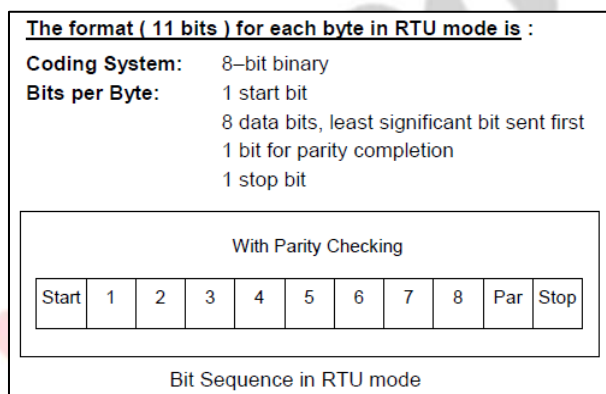


Figure 4-1 The format for each byte in RTU mode

4.4. Exception Responses

Except for broadcast messages, when a master device sends a query to a slave device it expects a normal response. One of four possible events can occur from the master's query:

- If the slave device receives the query without a communication error, and can handle the query normally, it returns a normal response.
- If the slave does not receive the query due to a communication error, no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query, but detects a communication error (parity, LRC, or CRC), no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query without a communication error, but cannot handle it (for example, if the request is to read a non-existent coil or register), the slave will return an exception response informing the master of the nature of the error.

The table below describes the exception codes supported by CyberPower Systems devices along with their possible causes.

Code	Name	Meaning
01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action for the slave. If a Poll Program Complete command was issued, this code indicates that no program function preceded it.
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the slave.
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for the slave.
04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the slave was attempting to perform the requested action.
06	SLAVE DEVICE BUSY	The slave is engaged in processing a long-duration program command. The master should retransmit the message later when the slave is free.

Table 4-1 Exception Codes Supported by CyberPower Systems Devices

4.5. CRC Generation

When RTU mode is used for character framing, the error checking field contains a 16-bit value implemented as two 8-bit bytes. The error check value is the result of a Cyclical Redundancy Check calculation performed on the message contents.

The CRC field is appended to the message as the last field in the message. When this is done, the low-order byte of the field is appended first, followed by the high-order byte. The CRC high-order byte is the last byte to be sent in the message.

The CRC value is calculated by the transmitting device, which appends the CRC to the message. The receiving device recalculates a CRC during receipt of the message, and compares the calculated value to the actual value it received in the CRC field. If the two values are not equal, an error results.

4.6. Addressing Rules

The Modbus addressing space comprises 256 different addresses. The addresses from 248 to 255 are reserved and the address 0 is broadcast address.

All CyberPower Systems products support broadcast request packets from the master. As its name implies, broadcasts allow all devices to receive and process the same command from the Modbus master. Broadcasts are triggered by setting the slave address to 0 (0x00).

0	From 1 to 247	From 248 to 255
Broadcast address	Slave individual addresses	Reserved

Table 4-2 Addressing rules

4.7. RTU Framing

In RTU mode, messages start with a silent interval of at least 3.5 character times. This is most easily implemented as a multiple of character times at the baud rate that is being used on the network. The first field then transmitted is the device address.

The allowable characters transmitted for all fields are hexadecimal 0–9, A–F. Networked devices monitor the network bus continuously, including during the 'silent' intervals. When the first field (the address field) is received, each device decodes it to find out if it is the addressed device.

Following the last transmitted character, a similar interval of at least 3.5 character times marks the end of the message. A new message can begin after this interval.

The entire message frame must be transmitted as a continuous stream. If a silent interval of more than 1.5 character times occurs before completion of the frame, the receiving device flushes the incomplete message and assumes that the next byte will be the address field of a new message.

Similarly, if a new message begins earlier than 3.5 character times following a previous message, the receiving device will consider it a continuation of the previous message. This will set an error, as the value in the final CRC field will not be valid for the combined messages.

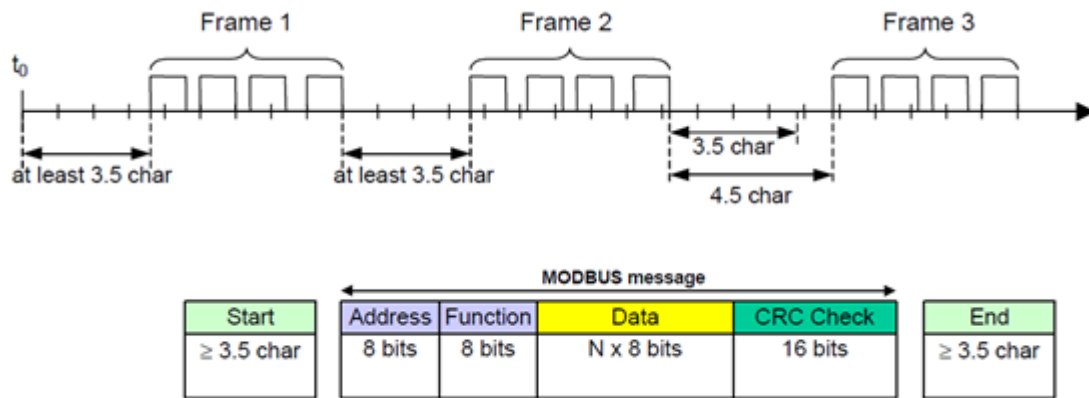


Figure 4-2 RTU Message Frame (1)

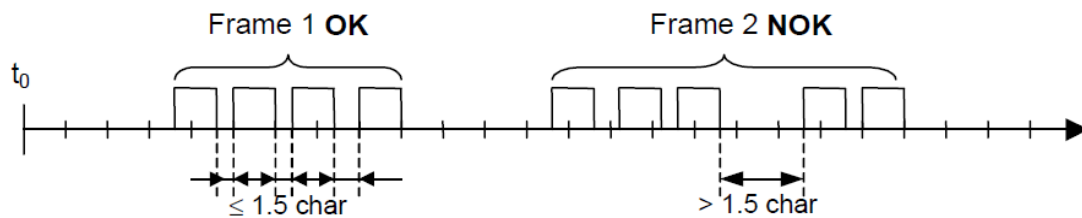


Figure 4-3 RTU Message Frame (2)

5. Supported Modbus Functions

5.1. Function 1 (01h): Read Coils

Description

Reads the ON/OFF status of discrete outputs.

Query

The query message specifies the starting coil and quantity of coils to be read.

Response

The coil status in the response message is packed as one coil per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the coil addressed in the query. The other coils follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes. If the returned coil quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Coils 6000h – 600Fh:

Request		Response	
Field Name	(Hex)	Field Name	(Hex)
Slave ID	01	Slave ID	01
Function	01	Function	01
Starting Address Hi	60	Byte Count	02
Starting Address Lo	00	Outputs status 6000-6007	00
Quantity of Outputs Hi	00	Outputs status 6008-600F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	23	CRC Hi	FC
CRC Hi	C6		

Table 5-1 example of a request to read Coils

5.2. Function 2 (02h): Read Discrete Inputs

Description

Reads the ON/OFF status of discrete inputs (1X references) in the slave.

Query

The query message specifies the starting input and quantity of inputs to be read.

Response

The input status in the response message is packed as one input per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the input addressed in the query. The other inputs follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes. If the returned input quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Discrete Inputs 5000h – 500Fh:

Request		Response	
Field Name	(Hex)	Field Name	(Hex)
Slave ID	01	Slave ID	01
Function	02	Function	02
Starting Address Hi	50	Byte Count	02
Starting Address Lo	00	Outputs status 5000-5007	00
Quantity of Outputs Hi	00	Outputs status 5008-500F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	68	CRC Hi	B8
CRC Hi	C6		

Table 5-2 example of a request to read Discrete Inputs

5.3. Function 3 (03h): Read Holding Registers

Description

Reads the binary contents of holding registers (4X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Holding Registers 0532h – 0533h:

Request		Response	
Field Name	(Hex)	Field Name	(Hex)
Slave ID	01	Slave ID	01
Function	03	Function	03
Starting Address Hi	05	Byte Count	04
Starting Address Lo	32	Register value Hi (0532)	07
No. of Registers Hi	00	Register value Lo (0532)	D0
No. of Registers Lo	02	Register value Hi (0533)	00
CRC Lo	65	Register value Lo (0533)	01
CRC Hi	08	CRC Lo	3B
		CRC Hi	7E

Table 5-3 example of a request to read Holding Registers

5.4. Function 4 (04h): Read Input Registers

Description

Reads the binary contents of input registers (3X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Input Registers 1000h – 1001h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	04	Function	04
Starting Address Hi	10	Byte Count	04
Starting Address Lo	00	Register value Hi (0532)	CC
No. of Registers Hi	00	Register value Lo (0532)	94
No. of Registers Lo	02	Register value Hi (0533)	06
CRC Lo	75	Register value Lo (0533)	0C
CRC Hi	0B	CRC Lo	86
		CRC Hi	9D

Table 5-4 example of a request to read Input Registers

5.5. Function 5 (05h): Write Single Coil

Description

Forces a single coil to either ON or OFF. When broadcast, the function forces the same coil reference in all attached slaves.

Query

The query message specifies the coil reference to be forced. The requested ON/OFF state is specified by a constant in the query data field. A value of FF 00 hex requests the coil to be ON. A value of 00 00 requests it to be OFF. All other values are illegal and will not affect the coil.

Response

The normal response is an echo of the query, returned after the coil state has been forced.

Here is an example of a request to write Coil 6000h ON:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	05	Function	05
Output Address Hi	60	Output Address Hi	60
Output Address Lo	00	Output Address Lo	00
Output Value Hi	FF	Output Value Hi	FF
Output Value Lo	00	Output Value Lo	00
CRC Lo	92	CRC Lo	92
CRC Hi	3A	CRC Hi	3A

Table 5-5 example of a request to write single Coil

5.6. Function 6 (06h): Write Single Register

Description

This function code is used to write a single holding register in a remote device. When broadcast, the function presets the same register reference in all attached slaves.

Query

The query message specifies the register reference to be preset. The requested preset value is specified in the query data field.

Response

The normal response is an echo of the request, returned after the register contents have been written.

Here is an example of a request to write Holding Register 0512h to 03E8h:

Request		Response	
Field Name	(Hex)	Field Name	(Hex)
Slave ID	01	Slave ID	01
Function	06	Function	06
Register Address Hi	05	Register Address Hi	05
Register Address Lo	12	Register Address Lo	12
Register Value Hi	03	Register Value Hi	03
Register Value Lo	E8	Register Value Lo	E8
CRC Lo	29	CRC Lo	29
CRC Hi	BD	CRC Hi	BD

Table 5-6 example of a request to write Holding Register

5.7. Function 16 (10h): Write Multiple Registers

Description

This function code is used to write a block of contiguous registers (1 to 123 registers). When broadcast, the function presets the same register references in all attached slaves.

Query

The requested written values are specified in the request data field. Data is packed as two bytes per register.

Response

The normal response returns the function code, starting address, and quantity of registers written.

Here is an example of a request to write three registers at 0532h to 07DCh, 0008h and 0014h:

Request		Response	
Field Name	(Hex)	Field Name	(Hex)
Slave ID	01	Slave ID	01
Function	10	Function	10
Starting Address Hi	05	Starting Address Hi	05
Starting Address Lo	32	Starting Address Lo	32
Quantity of Registers Hi	00	Quantity of Registers Hi	00
Quantity of Registers Lo	03	Quantity of Registers Lo	03
Byte Count	06	CRC Lo	21
Registers Value Hi	07	CRC Hi	0B
Registers Value Lo	DC		
Registers Value Hi	00		
Registers Value Lo	08		
Registers Value Hi	00		
Registers Value Lo	14		
CRC Lo	07		
CRC Hi	D0		

Table 5-7 example of a request to write three registers

6. CyberPower Systems Modbus Register Map

6.1. CyberPower Systems Modbus Data Formats

The MODBUS specification is not explicit on how to encode numbers other than 16 bit integers. Differences do exist between one manufacturer's implementation and another's. **This specification restricts values to the following standard data formats and it is difference to the formats in "SunSpec MODBUS Register Mappings".**

16 bit Integer Values

Values are stored in big--endian order per the MODBUS specification and consist of a single register. All integer values are documented a signed or unsigned. All signed values are represented using two's--compliment.

Modbus Register	1															
Byte	0								1							
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

32 bit Integer Values

32 bit integers are stored using two registers in big--endian order

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

64 bit Integer Values

64 bit integers are stored using four registers in big--endian order.

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

Modbus Register	3								4							
Byte	4				5				6				7			
Bits	47 ... 40				39 ... 32				63 ... 56				55 ... 48			

String Values

Store variable length string values in a fixed size register range using a NULL (0 value) to terminate or pad the string. For example, up to 16 characters can be stored in 8 contiguous registers as follows

EX : "EXAMPLE STRING!"

Modbus Register	1		2		3		4		5		6		7		8	
Byte	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Character	E	X	A	M	P	L	E	spc	S	T	R	I	N	G	!	NULL

NOT_IMPLEMENTED value: all registers filled with NULL or 0x0000.

Floating Point Values

Floating point values are 32 bits and encoded according to the IEEE 754 floating point standard.

Modbus Register	1		2		
Byte	0	1	2	3	
Bits	15 ... 8	7 ... 0	31	30 ... 24	23 ... 16
IEEE 754	Fraction least		sign	Exponent	Fraction

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6.2. Input Registers

6.2.1. Monitoring Parameters (Single Phase)

Modbus Address	Parameter	Format	Units /Scale	Description
1000 ~1001	Product ID	Uint32		CPSPV2000ETL : 6083918h => 101202200 CPSPV3000ETL : 609BFB8h => 101302200 CPSPV3600ETL : 60AAA18h => 101362200 CPSPV4000ETL : 60B4658h => 101402200 CPSPV5000ETL : 60CCCF8h => 101502200 EX : 07D1F4C8h => 131200200 1000 : F4C8h 1001 : 07D1h
1002	Inverter Rated Power	Uint16		AAB => AA*10^B, 007Bh => 123 => 12000W
1003	Peak Panel Power	Uint16		Connected solar power, AAB => AA*10^B, 007Bh => 123 => 12000W
1004	Installation Year	Uint16		Ex:2000.11.26 => 0x07D0, 0xFFFF => Not available,
1005	Installation Month	Uint16		Ex:2000.11.26 => 0x000B, 0xFFFF => Not available,
1006	Installation Day	Uint16		Ex:2000.11.26 => 0x001A, 0xFFFF => Not available,
1007	RS485 ID	Uint16		1 : ID=1 1 ~ 247 : ID=1~247
1008 ~100F	Serial Number	String		EX : EOE48010224 1008 : 454Fh, 1009 : 4534h, 100A : 3830h, 100B : 3130h, 100C : 3232h, 100D : 3400h, 100E : 0000h, 100F : 0000h,
1010 ~1017	Model Name	String		EX : EOE48010224 1010 : 454Fh, 1011 : 4534h, 1012 : 3830h, 1013 : 3130h, 1014 : 3232h, 1015 : 3400h, 1016 : 0000h, 1017 : 0000h,
1018	Processor1 FW Version	Uint16		Revision format: Hi byte: Major version Lo byte: Minor version 0x0A0B => V10.11 0xFFFF => Not available Date format: Hi byte: Year, starting from 2000 Lo byte: Week 082Dh => Y08W45 FFFFh => Not available Processor1: Communication Processor2: DSP Processor3: Redundant Processor4: Display Processor5: Ethernet Processor6: Flash memory
1019	Processor1 FW Date	Uint16		
101A	Processor1 HW Version	Uint16		
101B	Processor2 FW Version	Uint16		
101C	Processor2 FW Date	Uint16		
101D	Processor2 HW Version	Uint16		
101E	Processor3 FW Version	Uint16		
101F	Processor3 FW Date	Uint16		
1020	Processor3 HW Version	Uint16		
1021	Processor4 FW Version	Uint16		
1022	Processor4 FW Date	Uint16		
1023	Processor4 HW Version	Uint16		
1024	Processor5 FW Version	Uint16		
1025	Processor5 FW Date	Uint16		
1026	Processor5 HW Version	Uint16		
1027	Processor6 FW Version	Uint16		
1028	Processor6 FW Date	Uint16		

1029	Processor6 HW Version	Uint16		
102A	Grid code (Country)	Uint16		Refer to 'Appendix B: Grid codes and Languages Table'
102B	Custom Flag	Uint16		
102C	DC Connection Detection	Uint16		0000h : Auto detection 0001h : Manual
102D	DC Connection Status	Uint16		Refer to 'Appendix A: Data Structure'
102E	Language	Uint16		Refer to 'Appendix B: Countries and Languages Table'
102F	System Structure	Uint16		Refer to 'Appendix A: Data Structure'
1030	Inverter Status	Uint16		From DSP
1031	CPS_PV_Status	Uint16		Refer to 'Appendix A: Data Structure'
1032	Function Status Counter	Uint16		Bit 0-3: Event Counter (Add 1 when new event occur) Bit 4-7: Time Counter (Add 1 when changed time setting) Bit 8-11: ATE Counter (Add 1 when changed ATE setting) Bit 12-15: Reserved
1033	Reconnected Time	Uint16		
1034	Event Index Start	Uint16		
1035	Event Index End	Uint16		
1036	Input1 R Leakage	Uint16	1k ohm	
1037	Input2 R Leakage	Uint16	1k ohm	
1038 ~1039	Today kWh	Uint32	1 Wh	
103A ~103B	Today Runtime	Uint32	1 Sec	
103C ~103D	System Life kWh	Uint32	1 Wh	
103E ~103F	System Life Runtime	Uint32	1 Sec	
1040 ~1041	Input1 Today kWh	Uint32	1 Wh	
1042 ~1043	Input1 Today Runtime	Uint32	1 Sec	
1044 ~1045	Input1 Life kWh	Uint32	1 Wh	
1046 ~1047	Input1 Life Runtime	Uint32	1 Sec	
1048 ~1049	Input2 Today kWh	Uint32	1 Wh	
104A ~104B	Input2 Today Runtime	Uint32	1 Sec	
104C ~104D	Input2 Life kWh	Uint32	1 Wh	
104E ~104F	Input2 Life Runtime	Uint32	1 Sec	
1050	Input1 Volt (average)	Uint16	0.1 V	
1051	Input1 Curr (average)	Uint16	0.01 A	
1052	Input1 Watt (average)	Uint16	1 W	
1053	Input2 Volt (average)	Uint16	0.1 V	
1054	Input2 Curr (average)	Uint16	0.01 A	
1055	Input2 Watt (average)	Uint16	1 W	
1056	Bus Volt (Pos), (average)	Uint16	0.1 V	

1057	Bus Volt (Neg), (average)	Uint16	0.1 V	
1058	AC Volt (average)	Uint16	0.1 V	
1059	AC Curr (average)	Uint16	0.01 A	
105A	AC Watt (average)	Uint16	1 W	
105B	AC Freq (average)	Uint16	0.01 Hz	
105C	AC Volt (redundant) (average)	Uint16	0.1 V	
105D	AC Freq (redundant) (average)	Uint16	0.01 Hz	
105E	Temperature (Ambient)	Uint16	1 K	0xFFFF => Not available, 273K = 0°C, 373K = 100°C
105F	Temperature (Heatsink)	Uint16	1 K	0xFFFF => Not available, 273K = 0°C, 373K = 100°C
1060	Input1 Volt	Uint16	0.1 V	
1061	Input1 Curr	Uint16	0.01 A	
1062	Input1 Watt	Uint16	1 W	
1063	Input2 Volt	Uint16	0.1 V	
1064	Input2 Curr	Uint16	0.01 A	
1065	Input2 Watt	Uint16	1 W	
1066	Bus Volt (Pos)	Uint16	0.1 V	
1067	Bus Volt (Neg)	Uint16	0.1 V	
1068	AC Volt	Uint16	0.1 V	
1069	AC Curr	Uint16	0.01 A	
106A	AC Watt	Uint16	1 W	
106B	AC Freq	Uint16	0.01 Hz	
106C	AC Volt (redundant)	Uint16	0.1 V	
106D	AC Freq (redundant)	Uint16	0.01 Hz	
106E	Input1 R Leakage	Uint16	1k ohm	
106F	Input2 R Leakage	Uint16	1k ohm	
1070 ~108F	Reserved			
1090 ~1091	Grid code Support Table	Uint32		
1092 ~1093	Language Support Table	Uint32		
1094 ~1095	Reserved			
1096 ~1097	System Time Offset	Uint32	1 Sec	
1098 ~109F	Reserved			

6.2.2. Monitoring Parameters (Three Phase)

Modbus Address	Parameter	Format	Units /Scale	Description
1000 ~1001	Product ID	Uint32		CPSPV7000ETL : 60FDA38h => 101702200 CPSPV8000ETL : 61160D8h => 101802200 CPSPV9000ETL : 612E778h => 101902200 CPSPV10000ETL : 606B660h => 101103200 CPSPV12000ETL : 6070480h => 101123200 CPSPV18000ETL : 607EEE0h => 101183200 CPSPV20000ETL : 6083D00h => 101203200 CPSPV30000ETL : 609C3A0h => 101303200 CPSPV33000ETL : 60A38D0h => 101333200 CPSPV40000ETL : 60B4A40h => 101403200 EX : 060FDA38h => 101702200 1000 : DA38h 1001 : 060Fh
1002	Inverter Rated Power	Uint16		AAB => AA*10^B, 007Bh => 123 => 12000W
1003	Peak Panel Power	Uint16		Maximum input power, AAB => AA*10^B, 007Bh => 123 => 12000W
1004 ~1006	Reserved	Uint16		
1007	RS485 ID	Uint16		1 : ID=1 1 ~ 247 : ID=1~247
1008 ~100F	Serial Number	String		EX : EOE48010224 1008 : 454Fh, 1009 : 4534h, 100A : 3830h, 100B : 3130h, 100C : 3232h, 100D : 3400h, 100E : 0000h, 100F : 0000h,
1010 ~1017	Model Name	String		CPSPV7000ETL / CPSPV8000ETL / CPSPV9000ETL / CPSPV10000ETL / CPSPV12000ETL CPSPV18000ET / CPSPV20000ETL / CPSPV30000ETL / CPSPV33000ETL / CPSPV40000ETL EX : EOE48010224 1010 : 454Fh, 1011 : 4534h, 1012 : 3830h, 1013 : 3130h, 1014 : 3232h, 1015 : 3400h, 1016 : 0000h, 1017 : 0000h,
1018	Processor1 FW Version	Uint16		Revision format: Hi byte: Major version Lo byte: Minor version 0x0A0B => V10.11 0xFFFF => Not available Date format: Hi byte: Year, starting from 2000 Lo byte: Week 082Dh => Y08W45 FFFFh => Not available
1019	Processor1 FW Date	Uint16		
101A	Processor1 HW Version	Uint16		
101B	Processor2 FW Version	Uint16		
101C	Processor2 FW Date	Uint16		
101D	Processor2 HW Version	Uint16		
101E	Processor3 FW Version	Uint16		
101F	Processor3 FW Date	Uint16		
1020	Processor3 HW Version	Uint16		
1021	Processor4 FW Version	Uint16		
1022	Processor4 FW Date	Uint16		
1023	Processor4 HW Version	Uint16		
1024	Processor5 FW Version	Uint16		
1025	Processor5 FW Date	Uint16		

1026	Processor5 HW Version	Uint16		
1027	Processor6 FW Version	Uint16		
1028	Processor6 FW Date	Uint16		
1029	Processor6 HW Version	Uint16		
102A ~102C	Reserved			
102D	DC Connection Status	Uint16		Refer to 'Appendix A: Data Structure'
102E	Reserved			
102F	System Structure	Uint16		Refer to 'Appendix A: Data Structure'
1030	Inverter Status	Uint16		0 : Startup 4 : DC voltage low 8 : Grid volt/frequency error 10 : countdown 25 : MPPT (boot is OFF) 27 : MPPT (boost is ON) 50 : fault From DSP
1031	CPS_PV_Status	Uint16		Refer to 'Appendix A: Data Structure'
1032	Reserved			
1033	Reconnected Time	Uint16		Reconnected time (countdown time, second)
1034 ~1037	Reserved			
1038 ~1039	Today kWh	Uint32	1 Wh	
103A ~103B	Reserved			
103C ~103D	System Life kWh	Uint32	1 Wh	
103E ~103F	Reserved			
1040 ~1041	Input1 Today kWh	Uint32	1 Wh	
1042 ~1043	Reserved			
1044 ~1045	Input1 Life kWh	Uint32	1 Wh	
1046 ~1047	Reserved			
1048 ~1049	Input2 Today kWh	Uint32	1 Wh	
104A ~104B	Reserved			
104C ~104D	Input2 Life kWh	Uint32	1 Wh	
104E ~109F	Reserved			
10A0	AC_VoltageAN	Uint16	0.1 V	AC Voltage Phase A to N value
10A1	AC_VoltageBN	Uint16	0.1 V	AC Voltage Phase B to N value
10A2	AC_VoltageCN	Uint16	0.1 V	AC Voltage Phase C to N value
10A3	AC_CurrentA	Uint16	0.01 A	AC Phase A Current value
10A4	AC_CurrentB	Uint16	0.01 A	AC Phase B Current value
10A5	AC_CurrentC	Uint16	0.01 A	AC Phase C Current value
10A6	AC_PowerA	int16	1 W	AC Power value
10A7	AC_PowerB	int16	1 W	AC Power value
10A8	AC_PowerC	int16	1 W	AC Power value
10A9	AC_VoltageAB	Uint16	0.1 V	AC Voltage Phase AB value

10AA	AC_VoltageBC	Uint16	0.1 V	AC Voltage Phase BC value
10AB	AC_VoltageCA	Uint16	0.1 V	AC Voltage Phase CA value
10AC	AC_Frequency	Uint16	0.01 Hz	AC Frequency value
10AD ~10AF	Reserved			
10B0	DC1_Voltage	Uint16	0.1 V	DC Input 1 Voltage value
10B1	DC1_Current	Uint16	0.01 A	DC Input 1 Current value
10B2	DC1_Power	int16	1 W	DC Input 1 Power value
10B3	DC2_Voltage	Uint16	0.1 V	DC Input 2 Voltage value
10B4	DC2_Current	Uint16	0.01 A	DC Input 2 Current value
10B5	DC2_Power	int16	1 W	DC Input 2 Power value
10B6	BUS1_Voltage	Uint16	0.1 V	DC BUS 1 Voltage value
10B7	BUS2_Voltage	Uint16	0.1 V	DC BUS 2 Voltage value
10B8 ~10BB	Reserved			
10BC	Temperature (Amb)	Uint16		[°C] * 10, Ex: 266 = 26.6°C
10BD	Temperature 1	Uint16		[°C] * 10, Ex: 266 = 26.6°C
10BE	Temperature 2	Uint16		[°C] * 10, Ex: 266 = 26.6°C
10BF	Reserved			
10C0	CPS_PV_Status	Uint16		Refer to 'Appendix A: Data Structure'
10C1	Reconnected Time	Uint16	1 Sec	Reconnected time (countdown time, second), The same value in 1033h
10C2 ~10C3	I_Event_1	Uint32	Bitfield	
10C4 ~10C5	I_Event_2	Uint32	Bitfield	Event Flags (bits 32---63, reserved)
10C6	I_Event_1_Vendor	Uint16	Bitfield	Event Flags (E00 ~ E15, bit0 ~ bit15)
10C7	I_Event_2_Vendor	Uint16	Bitfield	Event Flags (E15 ~ E31, bit0 ~ bit15)
10C8	I_Event_3_Vendor	Uint16	Bitfield	Event Flags (W00 ~ W15, bit0 ~ bit15)
10C9	I_Event_4_Vendor	Uint16	Bitfield	Event Flags (F00 ~ F15, bit0 ~ bit15)
10CA	I_Event_5_Vendor	Uint16	Bitfield	Event Flags (F16 ~ F31, bit0 ~ bit15)
10CB	I_Event_6_Vendor	Uint16	Bitfield	Event Flags (F32 ~ F47, bit0 ~ bit15)
10CC	I_Event_7_Vendor	Uint16	Bitfield	Event Flags (F48 ~ F63, bit0 ~ bit15)
10CD	I_Event_8_Vendor	Uint16	Bitfield	Event Flags (F64 ~ F79, bit0 ~ bit15)
10CE ~10CF	Reserved			

6.2.3. kWh & Runtime Logs (**Single phase**)

Modbus Address	Parameter	Format	Units /Scale	Description
1400 ~14BF	AC Watt Average	Uint16	1 W / 1Wh	The AC watt average in every 450 from 00:00:00 to 23:59:59. 24 hour x 8 data = 192 1400h: 00:00:01 ~ 00:07:30, 1401h: 00:07:31 ~ 00:15:00 1402h: 00:15:01 ~ 00:22:30 : 14BEh: 23:45:01 ~ 23:52:30 14BFh: 23:52:31 ~ 00:00:00
14C0 ~14C1	Day-0 Wh	Uint32		(1) Daily energy in each day per month (2) Support Runtime
14C2 ~14C3	Day-0 Runtime	Uint32		
14C4 ~14C5	Day-1 Wh	Uint32		
14C6 ~14C7	Day-1 Runtime	Uint32		
:				
1538 ~1539	Day-30 Wh	Uint32		
153A ~153B	Day-30 Runtime	Uint32		
153C ~153D	Day-31 Wh	Uint32		
153E ~153F	Day-31 Runtime	Uint32		
:				
1540 ~1541	Month-0 Wh / kWh	Uint32		(1) Monthly energy in each month per year (2) Support Runtime
1542 ~1543	Month -0 Runtime	Uint32		
1544 ~1545	Month -1 Wh / kWh	Uint32		
1546 ~1547	Month -1 Runtime	Uint32		
:				
1568 ~1569	Month-10 Wh / kWh	Uint32		
156A ~156B	Month -10 Runtime	Uint32		
156C ~156D	Month -11 Wh / kWh	Uint32		
156E ~156F	Month -11 Runtime	Uint32		
:				

6.2.4. kWh & Runtime Logs (**Three phase**)

Modbus Address	Parameter	Format	Units /Scale	Description
1400 ~14BF	AC Watt Average	Uint16	1 W / 1Wh	The AC Wh in every 1 hour 1400h: Hour 0 (00:00 - 00:59) 1401h: Hour 1 (01:00 - 01:59) 1402h: Hour 2 (02:00 - 02:59)

				: 1416h: Hour 22 (22:00 - 22:59) 1417h: Hour 23 (23:00 - 23:59)
14C0 ~14C1	Day-0 Wh	Uint32		(1) Daily energy for recent 7 days (-0 means current day, i.e. today) (2) Do not support Runtime (3) Accumulating every 100Wh
14C2 ~14C3	Day-0 Runtime	Uint32		
14C4 ~14C5	Day-1 Wh	Uint32		
14C6 ~14C7	Day-1 Runtime	Uint32		
:				
1538 ~1539	Day-30 Wh	Uint32		
153A ~153B	Day-30 Runtime	Uint32		
153C ~153D	Day-31 Wh	Uint32		
153E ~153F	Day-31 Runtime	Uint32		
1540 ~1541	Month-0 Wh / kWh	Uint32		(1) Monthly energy for recent 12 month (-0 means current month) (2) Do not support Runtime (3) Accumulating every 1kWh
1542 ~1543	Month -0 Runtime	Uint32		
1544 ~1545	Month -1 Wh / kWh	Uint32		
1546 ~1547	Month -1 Runtime	Uint32		
:				
1568 ~1569	Month-10 Wh / kWh	Uint32		
156A ~156B	Month -10 Runtime	Uint32		
156C ~156D	Month -11 Wh / kWh	Uint32		
156E ~156F	Month -11 Runtime	Uint32		
1570 ~1571	Year-0 kWh	Uint32		(1) Yearly energy for recent 20 years (-0 means current year) (2) Do not support Runtime (3) Accumulating every 10kWh
1572 ~1573	Year-0 Runtime	Uint32		
1574 ~1575	Year-1 kWh	Uint32		
1576 ~1577	Year-1 Runtime	Uint32		
:				
15B8 ~15B9	Year-18 kWh	Uint32		
15BA ~15BB	Year-18 Runtime	Uint32		
15BC ~15BD	Year-19 kWh	Uint32		
15BE ~15BF	Year-19 Runtime	Uint32		

6.2.5. Event Log (Single phase, Refer to Appendix C for more detail)

Modbus Address	Parameter	Format	Units /Scale	Description
1800	Event -0 Index	Uint16		Event Index: 0(The latest) ~ 31(The older) Error Code: 0xABCD ° (1) AB means event type (3 types in ASCII) E code: 45h W code : 57h F code : 46h E code clear: C5h (Significant bit=1) W code clear: D7h (Significant bit=1) F code clear: C6h (Significant bit=1) (2) CD means event code Error: 0 - 31 Warning: 0 - 15 Fault: 0 - 79 (3) Event Time: Seconds since Jan/01/1970 00:00:00
1801	Event -0 Error Code	Uint16		
1802 ~1803	Event -0 Date/Time	Uint32	1 Sec	
1804	Event -1 Index	Uint16		
1805	Event -1 Error Code	Uint16		
1806 ~1807	Event -1 Date/Time	Uint32	1 Sec	
1808	Event -2 Index	Uint16		
1809	Event -2 Error Code	Uint16		
180A ~180B	Event -2 Date/Time	Uint32	1 Sec	
:				
1874	Event -29 Index	Uint16		
1875	Event -29 Error Code	Uint16		
1876 ~1877	Event -29 Date/Time	Uint32	1 Sec	
1878	Event -30 Index	Uint16		
1879	Event -30 Error Code	Uint16		
187A ~187B	Event -30 Date/Time	Uint32	1 Sec	
187C	Event -31 Index	Uint16		
187D	Event -31 Error Code	Uint16		
187E ~187F	Event -31 Date/Time	Uint32	1 Sec	

6.2.6. Event Log (Three phase, Refer to Appendix C for more detail)

Modbus Address	Parameter	Format	Units /Scale	Description
1800	Event -0 Index	Uint16		(1) Event Index: 0xFFFF (Not available) (2) Event Error Code: Refer to Appendix E (3) Event Time: Seconds since Jan/01/1970 00:00:00
1801	Event -0 Error Code	Uint16		
1802 ~1803	Event -0 Date/Time	Uint32	1 Sec	
1804	Event -1 Index	Uint16		
1805	Event -1 Error Code	Uint16		
1806 ~1807	Event -1 Date/Time	Uint32	1 Sec	
1808	Event -2 Index	Uint16		
1809	Event -2 Error Code	Uint16		
180A ~180B	Event -2 Date/Time	Uint32	1 Sec	
:				
1874	Event -29 Index	Uint16		
1875	Event -29 Error Code	Uint16		
1876 ~1877	Event -29 Date/Time	Uint32	1 Sec	
1878	Event -30 Index	Uint16		
1879	Event -30 Error Code	Uint16		
187A ~187B	Event -30 Date/Time	Uint32	1 Sec	

187C	Event -31 Index	Uint16		
187D	Event -31 Error Code	Uint16		
187E ~187F	Event -31 Date/Time	Uint32	1 Sec	

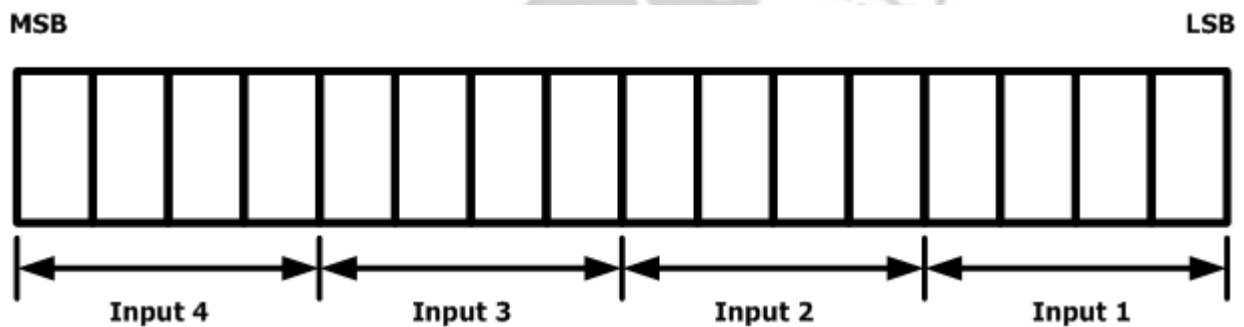
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Appendix A: Data Structure

※ CPS_PV_Status

STATUS_STARTING	0x0001	Starting up
STATUS_RECLOSURE	0x0002	Re-closure time
STATUS_ON	0x0004	Inverter is ON
STATUS_MPPT	0x0008	Tracking power point
STATUS_THROTTLED	0x0010	Forced power reduction
STATUS_REMOTE_OFF	0x0020	Remote Off
reserved	0x0040	
EVENT_INPUT_OVER_VOLT	0x0080	DC voltage above limit
EVENT_INPUT_UNDER_VOLT	0x0100	DC voltage under limit
EVENT_OVER_TEMP	0x0200	Over temperature
EVENT_OVER_FREQUENCY	0x0400	Frequency above limit
EVENT_UNDER_FREQUENCY	0x0800	Frequency under limit
EVENT_AC_OVER_VOLT	0x1000	AC voltage above limit
EVENT_AC_UNDER_VOLT	0x2000	AC voltage under limit
EVENT_GROUND_FAULT	0x4000	Ground (Insulation) fault
EVENT_FAULT	0x8000	One or more faults exist

※ DC Connection Status



0 : No support.

1 ~ 4 : This input is supported. The same number means parallel connection.

Others : Not acceptable.

For example :

0001h : Support one DC input only.

0021h : Support two DC inputs and they are separate connection.

0321h : Support three DC inputs and all of them are separate connection.

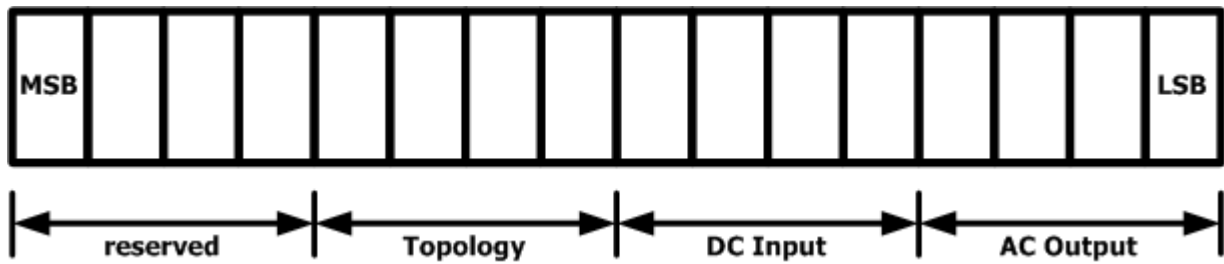
4321h : Support four DC inputs and all of them are separate connection.

0011h : Support two DC inputs and they are parallel connection.

1111h : Support four DC inputs and all of them are parallel connection.

2211h : Support four DC inputs. Input1 and input2 are parallel connection. Input3 and input4 are parallel connection.

※ System Structure



AC Output	0001h ~ 00Fh	0001h : Single Phase Inverter Model 0002h : Split Phase Inverter Model 0003h : Three Phase Inverter Model (Wye(Y)---Connect) 0004h : Three Phase Inverter Model (Delta(Δ)---Connect) Others : reserved
DC Input	0010h ~ 00F0h	0010h : Single DC Input 0020h : Two DC Inputs 0030h : Three DC Inputs 0040h : Four DC Inputs Others : reserved
Topology	0100h ~ 0F00h	0100h : Single BUS + DC/AC (TL) 0200h : Two BUS + DC/AC (TL) 0300h : DC/DC + Single BUS + DC/AC (TL) 0400h : DC/DC + Two BUS + DC/AC (TL) 0500h : DC/DC + Single BUS + DC/AC (TR) 0600h : DC/DC + Two BUS + DC/AC (TR) Others : reserved
Reserved	1000h ~ F000h	

Appendix B: Grid codes and Languages Table

1. Countries Table (Single phase)

Grid Code	Index	Corresponding bit in Grid codes Supported Registers (Input Registers : 1090~1091h)
Default	00h	Bit 0
Custom	01h	Bit 1
VDE-AR-N4105	02h	Bit 2
VDE0126-1-1	03h	Bit 3
UTE C15-712-1	04h	Bit 4
VDE0126-1-1/TW	05h	Bit 5
VDE0126-1-1/A1	06h	Bit 6
IEC61727and 62116	07h	Bit 7
IEC62116 PEA	08h	Bit 8
Reserved	09h	Bit 9
Reserved	0Ah	Bit 10
Reserved	0Bh	Bit 11
Reserved	0Ch	Bit 12
Reserved	0Dh	Bit 13
Reserved	0Eh	Bit 14
Reserved	0Fh	Bit 15
Reserved	10h	Bit 16
Reserved	11h	Bit 17
Reserved	12h	Bit 18
Reserved	13h	Bit 19
Reserved	14h	Bit 20
Reserved	15h	Bit 21
Reserved	16h	Bit 22
Reserved	17h	Bit 23
Reserved	18h	Bit 24
Reserved	19h	Bit 25
Reserved	1Ah	Bit 26
Reserved	1Bh	Bit 27
Reserved	1Ch	Bit 28
Reserved	1Dh	Bit 29
Reserved	1Eh	Bit 30
Reserved	1Fh	Bit 31

2. Countries Table (Three phase)

Grid Code	Index	Corresponding bit in Grid codes Supported Registers (Input Registers : 1090~1091h)
Default	00h	Bit 0
VDE0126-1-1	01h	Bit 1
VDE0126-1-1/A1	02h	Bit 2
VDE0126-1-1/TW	03h	Bit 3
PEA Thailand	04h	Bit 4
VFR 2014	05h	Bit 5
IEC61727+62116	06h	Bit 6

Reserved	07h	Bit 7
Reserved	08h	Bit 8
Reserved	09h	Bit 9
Reserved	0Ah	Bit 10
Reserved	0Bh	Bit 11
Reserved	0Ch	Bit 12
Reserved	0Dh	Bit 13
Reserved	0Eh	Bit 14
Reserved	0Fh	Bit 15
Reserved	10h	Bit 16
Reserved	11h	Bit 17
Reserved	12h	Bit 18
Reserved	13h	Bit 19
Reserved	14h	Bit 20
Reserved	15h	Bit 21
Reserved	16h	Bit 22
Reserved	17h	Bit 23
Reserved	18h	Bit 24
Reserved	19h	Bit 25
Reserved	1Ah	Bit 26
Reserved	1Bh	Bit 27
Reserved	1Ch	Bit 28
Reserved	1Dh	Bit 29
Reserved	1Eh	Bit 30
Reserved	1Fh	Bit 31

3. Languages Table (Single phase)

Language	Index	Corresponding bit in Language Supported Registers (Input Registers : 1092~1093h)
Default	00h	Bit 0
Reserved	01h	Bit 1
Reserved	02h	Bit 2
Reserved	03h	Bit 3
Reserved	04h	Bit 4
Reserved	05h	Bit 5
Reserved	06h	Bit 6
English	07h	Bit 7
French	08h	Bit 8
Reserved	09h	Bit 9
Reserved	0Ah	Bit 10
Reserved	0Bh	Bit 11
Reserved	0Ch	Bit 12
Reserved	0Dh	Bit 13
Reserved	0Eh	Bit 14
Reserved	0Fh	Bit 15
Reserved	10h	Bit 16
Reserved	11h	Bit 17
Reserved	12h	Bit 18
Reserved	13h	Bit 19
Reserved	14h	Bit 20
Reserved	15h	Bit 21
Reserved	16h	Bit 22
Reserved	17h	Bit 23

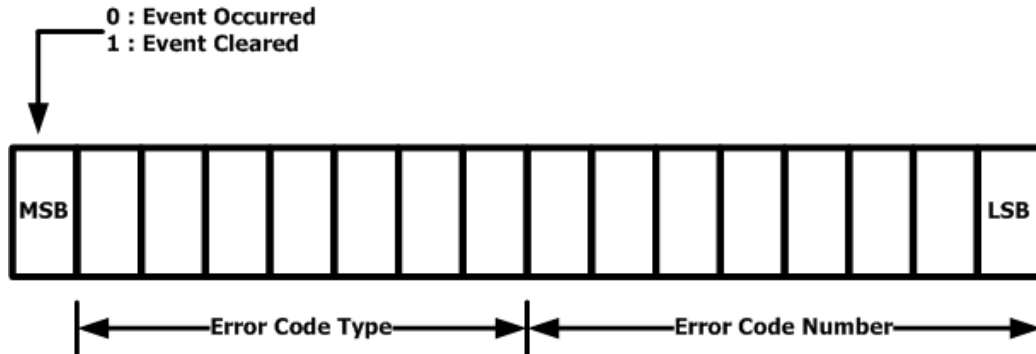
Reserved	18h	Bit 24
Reserved	19h	Bit 25
Reserved	1Ah	Bit 26
Reserved	1Bh	Bit 27
Reserved	1Ch	Bit 28
Reserved	1Dh	Bit 29
Reserved	1Eh	Bit 30
Reserved	1Fh	Bit 31

4. Languages Table (**Three phase**)

Language	Index	Corresponding bit in Language Supported Registers (Input Registers : 1092~1093h)
English	00h	Bit 0
French	01h	Bit 1
Chinese	02h	Bit 2
Reserved	03h	Bit 3
Reserved	04h	Bit 4
Reserved	05h	Bit 5
Reserved	06h	Bit 6
Reserved	07h	Bit 7
Reserved	08h	Bit 8
Reserved	09h	Bit 9
Reserved	0Ah	Bit 10
Reserved	0Bh	Bit 11
Reserved	0Ch	Bit 12
Reserved	0Dh	Bit 13
Reserved	0Eh	Bit 14
Reserved	0Fh	Bit 15
Reserved	10h	Bit 16
Reserved	11h	Bit 17
Reserved	12h	Bit 18
Reserved	13h	Bit 19
Reserved	14h	Bit 20
Reserved	15h	Bit 21
Reserved	16h	Bit 22
Reserved	17h	Bit 23
Reserved	18h	Bit 24
Reserved	19h	Bit 25
Reserved	1Ah	Bit 26
Reserved	1Bh	Bit 27
Reserved	1Ch	Bit 28
Reserved	1Dh	Bit 29
Reserved	1Eh	Bit 30
Reserved	1Fh	Bit 31

Appendix C: Error Code

1. Error Code Format for Single Phase



Error Code Type

E code: Error	45h
W code : Warning	57h
F code : Fault	46h

For example:

- "AC Over Frequency" exist : E01 => 4501h
- "Grid Quality" exist : E09 => 4509h
- "HW Bus OVR" clear : 8000h + F31 => C61Fh

2. Error Code Priority for Single Phase

Priority Level	Description	Notify Customer
0.x	Lowest priority. It's a normal condition	No
1.x	Warning. Inverter works with limitation.	YES
2.x	Grid voltage/frequency error.	YES
3.x	No Grid.	YES
4.x	Under temperature or reserved F code	YES
5.x	Protection or over specification	YES
6.x	Hardware fault	YES
7.x	Installation problem	YES
8.x	Highest priority. Firmware compatible problem	YES

※ This table indicates error code priorities in N.X. The N means its group priority level. The X means its priority in this group. A lower value has a lower priority and a higher value has a higher priority.

3. "E" Error Code Table for Single Phase

Error Code	Name	Hex Value	Priority	Description
E00	No Grid	4500	3.0	AC Phase A shutdown
E01	AC Over Frequency (A)	4501	2.24	AC Phase A Frequency above limit
E02	AC Under Frequency (A)	4502	2.21	AC Phase A Frequency under limit
E03	AC Over Volt (A)	4503	2.30	AC Phase A Voltage above limit
E04	AC Under Volt (A)	4504	2.27	AC Phase A Voltage under limit
E05	E05	4505	2.17	Reserved
E06	E06	4506	2.16	Reserved

E07	E07	4507	2.15	Reserved
E08	E08	4508	2.14	Reserved
E09	Grid Quality	4509	2.18	reserved
E10	E10	450A	2.13	Reserved
E11	AC Over Frequency (B)	450B	2.23	AC Phase B Frequency above limit
E12	AC Under Frequency (B)	450C	2.20	AC Phase B Frequency under limit
E13	AC Over Volt (B)	450D	2.29	AC Phase B Voltage above limit
E14	AC Under Volt (B)	450E	2.26	AC Phase B Voltage under limit
E15	AC Over Frequency (C)	450F	2.22	AC Phase C Frequency above limit
E16	AC Under Frequency (C)	4510	2.19	AC Phase C Frequency under limit
E17	AC Over Volt (C)	4511	2.28	AC Phase C Voltage above limit
E18	AC Under Volt (C)	4512	2.25	AC Phase C Voltage under limit
E19	E19	4513	2.12	Reserved
E20	E20	4514	2.11	Reserved
E21	E21	4515	2.10	Reserved
E22	E22	4516	2.9	Reserved
E23	E23	4517	2.8	Reserved
E24	E24	4518	2.7	Reserved
E25	E25	4519	2.6	Reserved
E26	E26	451A	2.5	Reserved
E27	E27	451B	2.4	Reserved
E28	E28	451C	2.3	Reserved
E29	E29	451D	2.2	Reserved
E30	E30	451E	2.1	Reserved
E31	E31	451F	2.0	Reserved

4. "W" Error Code Table for Single Phase

Error Code	Name	Hex Value	Priority	Description
W00	Input1 Under Volt	5700	0.3	Input1 DC Voltage under limit
W01	Input2 Under Volt	5701	0.2	Input2 DC Voltage under limit
W02	Input3 Under Volt	5702	0.1	Input3 DC Voltage under limit
W03	Input4 Under Volt	5703	0.0	Input4 DC Voltage under limit
W04	Input1 Over Volt	5704	7.4	Input1 DC Voltage above limit
W05	Input2 Over Volt	5705	7.3	Input2 DC Voltage above limit
W06	Input3 Over Volt	5706	7.2	Input3 DC Voltage above limit
W07	Input4 Over Volt	5707	7.1	Input4 DC Voltage above limit
W08	W08	5708	1.6	Reserved
W09	W09	5709	1.5	Reserved
W10	HW Fan	570A	1.7	Fan Failed
W11	W11	570B	1.4	Reserved
W12	W12	570C	1.3	Reserved
W13	W13	570D	1.2	Reserved
W14	W14	570E	1.1	Reserved
W15	W15	570F	1.0	reserved

5. "F" Error Code Table for Single Phase

Error Code	Name	Hex Value	Priority	Description
F00	F00	4600	4.32	Reserved
F01	AC Current DC	4601	5.4	AC Phase A Current DC Component

	Component above limit (A)			above limit
F02	AC Current DC Component above limit (B)	4602	5.3	AC Phase B Current DC Component above limit
F03	AC Current DC Component above limit (C)	4603	5.2	AC Phase C Current DC Component above limit
F04	F04	4604	4.31	Reserved
F05	Under Temp	4605	4.33	Temperature under limit
F06	NTC1 Over Temp	4606	5.24	NTC1 Temperature over limit
F07	HW NTC1 Failure	4607	6.3	Temperature sensor (NTC1) error, or heating circuit failure
F08	NTC2 Over Temp	4608	5.23	NTC2 Temperature over limit
F09	HW NTC2 Failure	4609	6.2	Temperature sensor (NTC2) error, or heating circuit failure
F10	NTC3 Over Temp	460A	5.22	NTC3 Temperature over limit
F11	HW NTC3 Failure	460B	6.1	Temperature sensor (NTC3) error, or heating circuit failure
F12	NTC4 Over Temp	460C	5.21	NTC4 Temperature over limit
F13	HW NTC4 Failure	460D	6.0	Temperature sensor (NTC4) error, or heating circuit failure
F14	Firmware are not compatible	460E	8.0	The firmwares are not compatible
F15	HW DSP ADC1	460F	6.18	DSP ADC1 sensor error or circuit failure
F16	HW DSP ADC2	4610	6.17	DSP ADC2 sensor error or circuit failure
F17	HW DSP ADC3	4611	6.16	DSP ADC3 sensor error or circuit failure
F18	HW DSP ADC4	4612	6.15	DSP ADC4 sensor error or circuit failure
F19	HW Red ADC1	4613	6.14	Redundant MCU ADC1 sensor error or circuit failure
F20	HW Efficiency	4614	6.13	Input/output voltage/current sensor error or circuit failure
F21	HW FAN	4615	6.4	Fan circuit failure
F22	HW COMM1	4616	6.5	The serial communication port between DSP and comm. MCU is failure
F23	HW COMM2	4617	6.6	The serial communication port between DSP and redundant MCU is failure
F24	Ground Current	4618	5.8	Residual current is above limit
F25	RCMU Fail	4619	6.7	Residual current sensor error or circuit failure
F26	Insulation	461A	7.0	Insulation resistance to Earth is under limit
F27	F27	461B	4.30	Reserved
F28	HW Relay Short	461C	6.12	Relay or drive circuit failure
F29	HW Relay Open	461D	6.11	Relay or drive circuit failure
F30	Bus Unbalance	461E	5.14	Bus1 and Bus2 voltage are not balance
F31	HW Bus OVR	461F	5.13	Bus1 voltage above limit or circuit failure
F32	HW Bus UVR	4620	5.1	Bus1 voltage under limit or circuit failure
F33	HW Bus OVR	4621	5.12	Bus2 voltage above limit or circuit failure
F34	HW Bus UVR	4622	5.0	Bus2 voltage under limit or circuit failure
F35	HW Bus OVR	4623	5.11	Bus voltage above limit or circuit failure
F36	AC Over Current (A) transient	4624	5.7	AC Phase A Current above limit (transient)
F37	AC Over Current (A)	4625	5.20	AC Phase A Current above limit
F38	AC Over Current (B) transient	4626	5.6	AC Phase B Current above limit (transient)

F39	AC Over Current (B)	4627	5.19	AC Phase B Current above limit
F40	AC Over Current (C) transient	4628	5.5	AC Phase C Current above limit (transient)
F41	AC Over Current (C)	4629	5.18	AC Phase C Current above limit
F42	HW CT Fail (A)	462A	6.10	AC Phase A Current sensor or circuit failure
F43	HW CT Fail (B)	462B	6.9	AC Phase B Current sensor or circuit failure
F44	HW CT Fail (C)	462C	6.8	AC Phase C Current sensor or circuit failure
F45	HW AC OCR	462D	5.15	Hardware AC over current protection
F46	F46	462E	4.29	Reserved
F47	F47	462F	4.28	Reserved
F48	F48	4630	4.27	Reserved
F49	F49	4631	4.26	Reserved
F50	F50	4632	4.25	Reserved
F51	F51	4633	4.24	Reserved
F52	F52	4634	4.23	Reserved
F53	F53	4635	4.22	Reserved
F54	F54	4636	4.21	Reserved
F55	F55	4637	4.20	Reserved
F56	F56	4638	4.19	Reserved
F57	F57	4639	4.18	Reserved
F58	F58	463A	4.17	Reserved
F59	F59	463B	4.16	Reserved
F60	Input1 Over Current	463C	5.17	Input1 Current above limit
F61	Input2 Over Current	463D	5.16	Input2 Current above limit
F62	F62	463E	4.15	Reserved
F63	F63	463F	4.14	Reserved
F64	F64	4640	4.13	Reserved
F65	F65	4641	4.12	Reserved
F66	F66	4642	4.11	Reserved
F67	F67	4643	4.10	Reserved
F68	F68	4644	4.9	Reserved
F69	F69	4645	4.8	Reserved
F70	Input1 Over Current transient	4646	5.10	Input1 Current above limit (transient)
F71	Input2 Over Current transient	4647	5.9	Input2 Current above limit (transient)
F72	F72	4648	4.7	Reserved
F73	F73	4649	4.6	Reserved
F74	F74	464A	4.5	Reserved
F75	F75	464B	4.4	Reserved
F76	F76	464C	4.3	Reserved
F77	F77	464D	4.2	Reserved
F78	F78	464E	4.1	Reserved
F79	F79	464F	4.0	Reserved

※ **Display its error code (ex: F76) in event log if it is a reserved code.**

※ **Don't display '(A)' in single phase inverter.**

For example E01 in single phase inverter : AC Over Frequency

For example E01 in three phase inverter : AC Over Frequency (A)

For example F42 in single phase inverter : HW CT Fail

For example F42 in three phase inverter : HW CT Fail (A)

6. Error Code Table for Three Phase

Error Code	Name	Hex Value	Priority	Description
101	Comm fault	0x0065	6.5	The serial communication failure from DSP to COM
121	Comm fault	0x0079	6.6	The serial communication failure from COM to DSP
122	BusVFault	0x007A	5.1	Bus voltage is out of range
117	RelayFault	0x0075	6.1	Relay failure
119	GFCIDamage	0x0077	6.7	Residual current sensor error or circuit Failure
125	PVISO	0x007D	7.0	Insulation resistance to Earth is under limit
126	GFCIHigh	0x007E	5.8	Residual current is above limit
127	DCIHigh	0x007F	5.4	DC Component above limit
128	PVHigh	0x0080	7.4	Input1 DC Voltage above limit
129	GridVFault	0x0081	3.0	AC voltage is out of range
130	GridFFault	0x0082	2.2	AC frequency is out of range
104	Firmware are not compatible	0x0068	8.0	The firmwares are not compatible
100	HW FAN	0x0064	6.4	Fan circuit failure

※ Error code 104 and 100 will not be recorded in the event log. (Just for display warning)